digis²
Enabling Smarter Networks

ITI Ismailia Branch

NetFix AI

AI-Powered Telecom Intelligence Platform

DIGISsquared Graduation Project
ITI Ismailia Branch — Intake 46

Drive Test Intelligence

- Throughput anomaly detection
- Root cause analysis (RCA)
- KPI and map-based insights
- AI-generated diagnostic reporting

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Planning Intelligence

- Site and sector planning
- Expected coverage preview
- PCI clash/confusion validation
- AI assistant for planning review

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One unified AI portal for planning intelligence, drive test intelligence, guided analysis, and smarter optimization decisions.

Presented by DIGIS Graduation Project Team

NetFix AI: Drive Test Intelligence

AI-powered anomaly detection, root-cause analysis, and guided optimization insights

PURPOSE

Detect performance issues earlier, prioritize what matters, and connect each anomaly to the most likely root cause.

OUTCOME

Surface top anomalies, explain likely causes, support decisions with KPI evidence, and present everything clearly on one dashboard.

1. DRIVE TEST DATA INPUT

- Drive test data upload
- KPI and location context
- Cell / sector level observations

2. ANOMALY DETECTION FAMILIES

- Statistical detection
- Rule / threshold detection
- Time-series behavior
- ML-based detection
- Pattern / peer comparison

Focus: Throughput anomalies
Used together to identify and rank throughput anomalies.

3. ROOT CAUSE ANALYSIS (RCA)

The system links each throughput anomaly to the most probable cause family using KPI evidence and context.

Throughput / Performance Degradation

- **Coverage** — weak serving / low RSRP
- **Interference** — poor SINR / CQI degradation
- **Congestion** — utilization / allocation efficiency
- **Mobility** — RACH / handover issues
- **Carrier Aggregation** — weak secondary cell / CA inactive
- **MIMO** — poor spatial multiplexing

4. AI RCA REPORT & PORTAL VIEW

An LLM connects the detected throughput anomaly with the most likely RCA and presents a documented report supported by graphs, KPIs, and cell ranking.

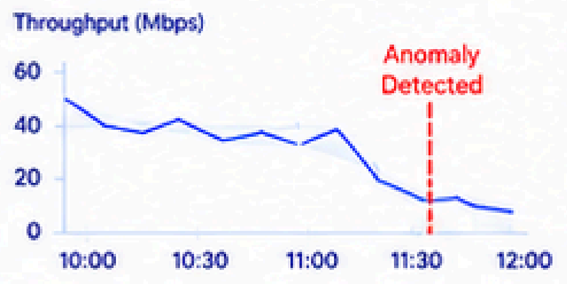
Top Throughput Anomalies

Rank	Cell / Site	Severity	Impact
1	Cell 10123	High	High
2	Cell 10087	High	High
3	Cell 10564	Medium	Medium
4	Cell 10211	Medium	Medium
5	Cell 10345	Low	Low

Most Probable RCA

Congestion
Utilization / Allocation Efficiency

KPI Trend (Throughput)



KPI Snapshot (At Anomaly)

DL Throughput	RSRP	SINR	PRB Utilization
8.2 Mbps	-112 dBm	2.4 dB	92% High

Cell Ranking (By Impact)

Rank	Cell ID	Impact Score	Anomaly
1	10123		High
2	10087		High
3	10564		Medium
4	10211		Medium
5	10345		Low

AI RCA Report

Throughput degradation is most likely caused by congestion due to high utilization and low RB availability. Coverage and interference are within acceptable limits. Recommended actions are listed below.

View Full Report →

One AI-powered portal for throughput anomaly detection, RCA insight, cell ranking, and guided optimization actions.

NetFix AI: Planning Intelligence

AI-powered site planning, validation, PCI analysis, and guided optimization insights

PURPOSE

Support smarter site planning, validate design choices early, and reduce configuration risks before deployment.

OUTCOME

Show expected coverage, detect PCI clashes and confusion, compare against pre-planned areas, and present planning insights clearly in one portal.

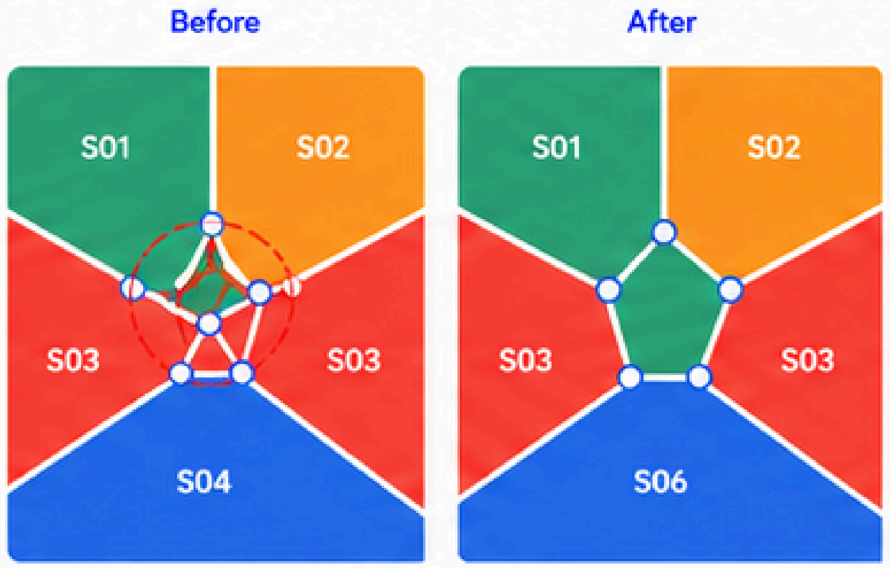
1. PLANNING DATA INPUT

- Site / sector planning data
- Existing reference plan
- Area and configuration context

Used as the foundation for planning validation and expected coverage analysis.

2. SITE PLANNING & EXPECTED COVERAGE

The system organizes planned sites/sectors and previews expected serving areas.



- Voronoi-style expected coverage view
- Site / sector grouping
- Supports planning review before rollout

3. PLANNING VALIDATION & PCI CHECKS

The platform validates the plan and highlights likely design issues.

Planning Validation

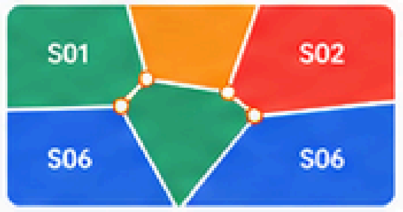
- PCI clash detection
- PCI confusion detection
- Compare with pre-planned area
- Configuration / layout checks

Helps identify planning risks before deployment.

4. AI ASSISTANT & PORTAL VIEW

An AI assistant explains findings, answers planning questions, and presents results in a clear report within the portal.

Expected Coverage



PCI Findings

Clash Risk	High
Confusion Risk	Medium

Plan Validation Summary

Coverage Completeness	Good
PCI Clash Risk	High
PCI Confusion Risk	Medium
Configuration Checks	Pass
Pre-planned Area Match	Good

AI Assistant

The plan shows high clash risk near the central area. Consider adjusting S02 azimuth or tilt to improve separation.

Recommended Actions

→ Adjust S02 azimuth / tilt to reduce clash
→ Reduce overlap between S01 and S05
→ Recheck coverage on the northern edge

View Full Report →

One AI-powered portal for site planning, expected coverage validation, PCI risk detection, and guided planning decisions.